INCH-POUND

MIL-C-24643/42A
14 March 1994
SUPERSEDING
MIL-C-24643/42
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS1SU (INCLUDING VARIATION LS1SUA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in two variations: LS1SU (unarmored) and LS1SUA (armored).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

	First	- Copper conductor, tin coated, ASTM B 286, size 20-7, maximum diameter 0.039 inch and AWG no. 18, class B stranding, 7 by 0.0152 inch, maximum diameter 0.050 inch. For LS1SU-36 32 size 20-7 conductors and 4 AWG no. 18 conductors. For LS1SU-60, 60 size 20-7 conductors.
	Second	- Separator may be used at manufacturer's option where required to provide free-stripping insulation.
*	Third	- Crosslinked polyethylene insulation. Nominal wall thickness 0.017 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements.
*	Fourth	- Braided shield of tin coated copper. Minimum coverage of 85 percent.
*	Fifth	- Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, outer tape sealed, an alternate shield insulation may consist of one tape, type G of MIL-I-631, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape. The four size AWG no. 18 conductors shall be numbers 1 through 4.

AMSC N/A FSC 6145
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Construction (non-watertight) (continued)

* Sixth - The required number of single (see table I) cabled around a center filler with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with no. 1, from the center outward. Fillers shall be used as necessary to form a firm well rounded cable.

Seventh - Binder tape, applied helically with overlap.

Construction variant for type LS1SU

* Eighth - Cross-linked polyolefin jacket. (See table I for thickness).

Construction variant for type LS1SUA

* Eighth - Cross-linked polyolefin jacket. (See table I for thickness).

* Ninth - Braided metal armor

* TABLE I. Details.

Military 1/ part no. M24643/42	Size 2/	Con-ductor size AWG no.	Dia over completed conductor (nominal) (inch)	Cable jacket thickness min avg (inch)	Overall diameter (inches)			
					TYPE LS1SU		TYPE LS1SUA	
					min	max	min	max
-01UN	36	18	0.127	0.075	0.910	0.985	0.960	1.035
-02UN	60	20-7	.115	0.100	1.219	1.310	1.260	1.360

1/ Unarmored. When armor option required, see MIL-C-24643 for configuration of part number

2/ Type LS1SU and LS1SUA.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor Resistance - ohms/1000 feet at 25°C, maximum ..	
ASTM B 286, size 20-7	10.93
Size AWG no. 18 (class B)	7.05
Voltage withstand - volts, root mean square, minimum	
Conductor to shield	2200
Shield to shield	500
Insulation resistance - Megohms/1000 feet minimum	
Conductor to shield	500
* Component shield to shield	100
* Conductor and shield continuity	No failure

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EXAMINATION AND TESTS (continued):

	Requirements
Group A:	
Visual and dimensional	No failure
Capacitance - at 1 kilohertz (KHz), pF/ft, maximum	72
Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Cold bending, cable - mandrel diameter 12 times cable diameter	No damage
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	-
* Tensile strength - percent of unaged, minimum	50
* Elongation - percent of unaged, minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
* Permanence of printing (jacket) - (type LS1SU only) cycles, minimum	125
Armor (type LS1SUA only) - conformance to material, construction and coverage	No failure
Shield - conformance to material, construction and coverage	No failure
Group D:	
* Flame propagation (cable)	No failure

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QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable) (125 ± 5°C.)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
* Shield insulation	2
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
* Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

All sizes 1000 feet (nominal)

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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